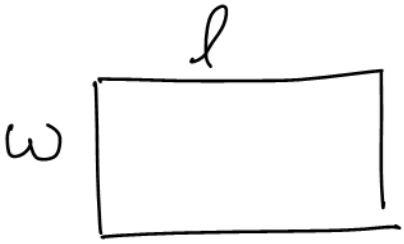
Goal: Use the tools used to find maximums and minimums in applications.

Main Idea: Given a situation, find the "optimal" value (usually a maximum or minimum).

Example 1. A farmer is building a rectangular pig pen and has 400′ of fencing to work with. What should the dimensions of the pen be to make the area as large as possible?

The function we want to optimize is called the objective function. What is it for this example?



Objective Maximize $A_{(l,w)} = lw$

In order to reduce the number of variables down to 1 and/or guarantee a max/min, we need more conditions called constraints. Are there any in this example? "have [we] 400' of fencing"

$$2w + 2l = 400$$
$$2l = 400 - 2w$$
$$l = 200 - w$$

We can now finish the problem (by finding the absolute max).

Rewrite A solely as a fcn of w →

$$A(l, w) = lw$$
$$A(w) = (200 - w)w$$

Find the possible locations for the absolute max:

$$A(w) = 200w - w^2$$
$$\frac{dA}{dw} = 200 - 2w \overset{\text{set}}{=} 0$$

$$200 = 2w$$
$$w = 100$$

Verify A has an absolute max at $w=100$:

$$\frac{d^2A}{dw^2} = -2 < 0$$

A has a rel. max at $w=100$ by the 2nd $\frac{d}{dx}$ test

Rel. max at $w=100$

$+$

Only One Crit. Num $\Longrightarrow$ Abs. max at $w = 100$

Answer the question:

I.e. we need to find l when $w = 100$

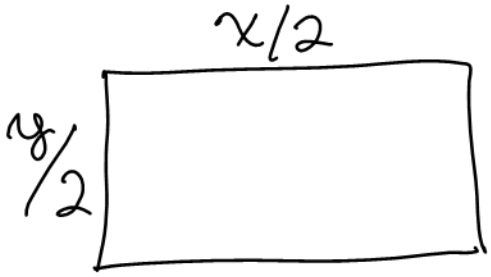
$$l = 200 - w$$
$$l|_{w=100} = 200 - 100 = 100'$$

Therefore, the area is maximized when it is a square with side lengths of $100'$.

$$x + y = S$$

xy maximized

Example 2. Suppose that two numbers x and y add up to $S > 0$. When is their product at a maximum?



Obj: Max $A_{(x,y)} = xy$

Given: $x + y = S$

$y = S - x$

$A(x,y) = xy$

$A(x) = x(S-x)$

$A(x) = Sx - x^2$

$\frac{dA}{dx} = S - 2x \overset{\text{set}}{=} 0$

$x = \frac{S}{2}$

Verify it is the loc. of the max.

$\frac{d^2A}{dx^2} = -2 < 0$

Since there is only 1 crit. num., A has an abs. max. at $x = \frac{S}{2}$ by the 2nd $\frac{d}{dx}$ test.

Find y: $y = S - x$

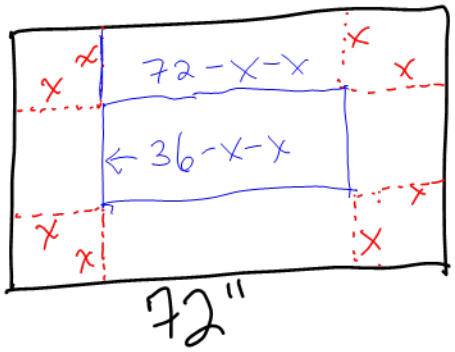
$y\big|_{\frac{S}{2}} = S - \frac{S}{2} = \frac{S}{2}$

I.e. their product is maximized when

$x = y = \frac{S}{2}$

36 inches

Example 3. A $36'' \times 72''$ piece of cardboard is cut from each corner and the sides are folded up to create a box with an open top. How far from the edge should the cut be to maximize the volume?



Obj Maximize

$$V(x) = (36-2x)(72-2x)x$$

Given: $0 \le x \le 18 \rightarrow x \in [0,18]$

$$V(x) = 2592x - 216x^2 + 4x^3$$

$$\frac{dV}{dx} = 2592 - 432x + 12x^2 \overset{\text{set}}{=} 0$$

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$\Rightarrow x = 18 \pm 6\sqrt{3}$$

$x \in [0,18]$

$$\Rightarrow x = 18 - 6\sqrt{3} \approx 7.61$$

x	0		$18-6\sqrt{3}$		18
$(36-2x)(72-2x)x$	0		>0		0

V has an abs. max. at $x = 18-6\sqrt{3}$.

Therefore, we only need to cut roughly 7.61" away from the edge to maximize.

Example 4. You are making a rectangular garden and you have 50′ of fencing to work with. To make a bigger garden, you build it right next to your house. How large can the garden be?



Obj Maximize $A = lw$

Given: $2w + l = 50$

$l = 50 - 2w$

$A = lw$

$A(w) = (50 - 2w)w$

$A(w) = 50w - 2w^2$

$\frac{dA}{dw} = 50 - 4w \overset{\text{set}}{=} 0$

$w = \frac{50}{4} = \frac{25}{2}$

$\frac{d^2A}{dw^2} = -4 < 0$ + only 1 Crit. Num $\Longrightarrow$ Abs. Max at $w = \frac{25}{2}$

To answer the question, we need to compute $A\left(\frac{25}{2}\right)$

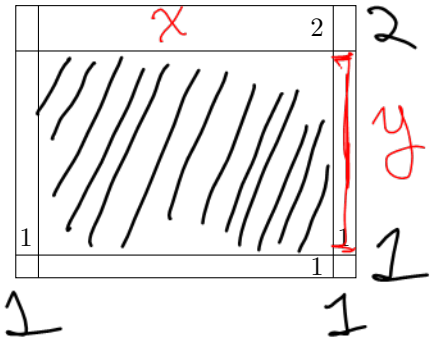
$$A\left(\frac{25}{2}\right) = \left(50 - 2\left(\frac{25}{2}\right)\right)\left(\frac{25}{2}\right)$$
$$= 25\left(\frac{25}{2}\right)$$
$$= \frac{25^2}{2} = 312.5 \text{ ft}^2$$

The maximum area is 312.5 ft^2.

Example 5. A poster board has an area of $180\,\text{cm}^2$ and has margins as shown in the diagram below. What is the largest printable area?

Obj:

Max $A(x,y) = xy$



Given:

$$(x+1+1)(y+1+2) = 180 \text{ AND } x, y \geq 0$$

$$(x+2)(y+3) = 180$$

$$x+2 = \frac{180}{y+3}$$

$$x = \frac{180}{y+3} - 2$$

$$A(x,y) = xy$$

$$A(y) = \left[\frac{180}{y+3} - 2\right] y$$

$$A(y) = \frac{180y}{y+3} - 2y$$

$$\frac{dA}{dy} = \frac{180(y+3) - 180y}{(y+3)^2} - 2$$

$$= \frac{180y + 540 - 180y}{(y+3)^2} - 2$$

$$\frac{dA}{dy} = \frac{540}{(y+3)^2} - 2 \overset{\text{set}}{=} 0$$

$$\frac{540}{(y+3)^2} = 2$$

$$\frac{540}{2} = (y+3)^2$$

$$(y+3)^2 = 270$$

$$y+3 = \pm 3\sqrt{30}$$

$y \geq 0 \longrightarrow y = -3 + 3\sqrt{30}$

$$A(-3+3\sqrt{30}) \approx 120.3 \text{ cm}^2$$